Folios — Architecture

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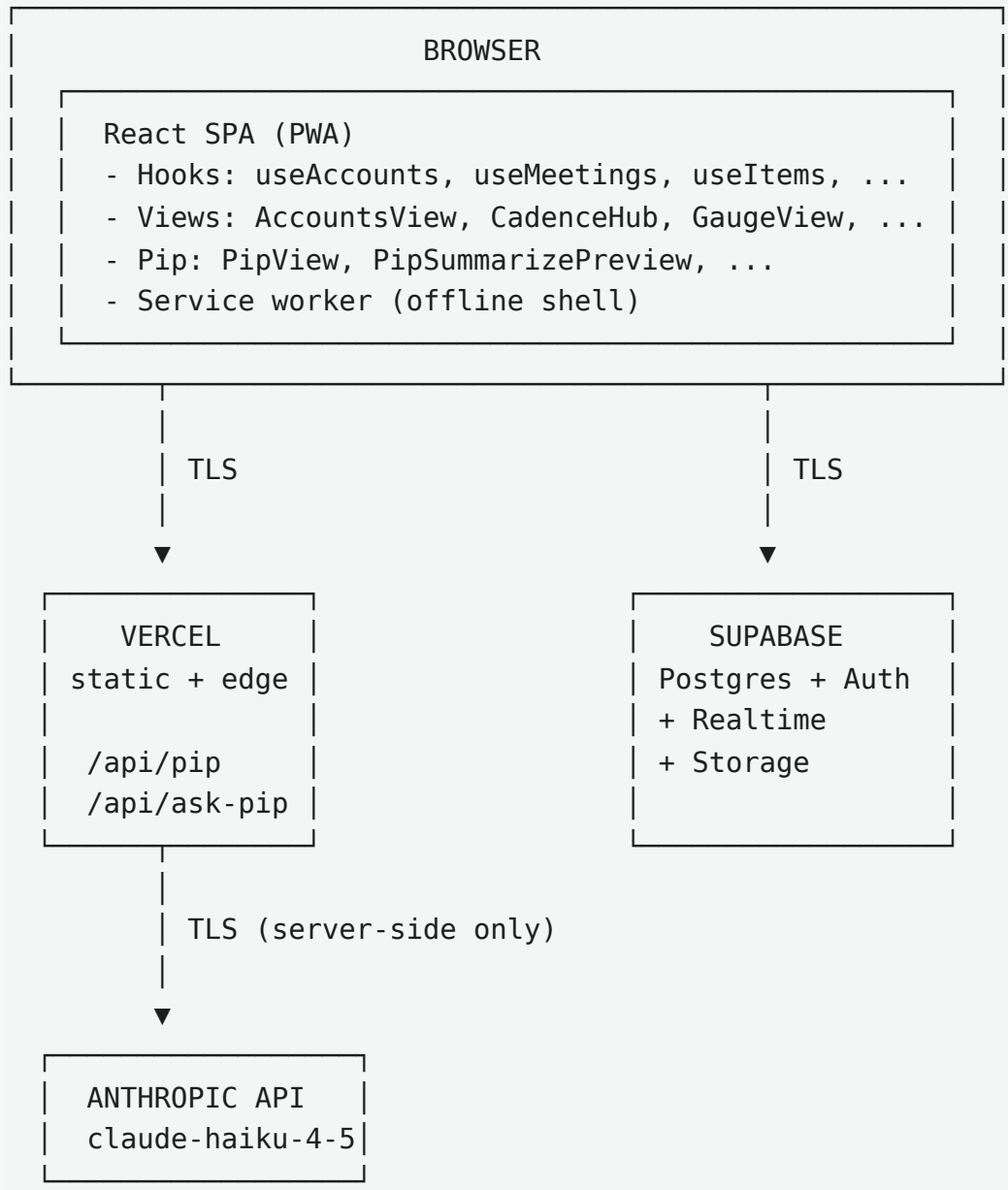
This document describes how Folios is built. Intended for engineering reviewers, technical due diligence, or anyone evaluating the system beyond the product surface.

Stack at a glance

Layer	Technology
Frontend	React 18 + Vite 5 (SPA)
Distribution	PWA (vite-plugin-pwa, Workbox)
Hosting	Vercel (static + serverless functions)
Database	Supabase Postgres 15
Auth	Supabase Auth (email/password + TOTP MFA)
Realtime	Supabase Realtime (Postgres LISTEN/NOTIFY)
AI inference	Anthropic API (claude-haiku-4-5-20251001, Sonnet for synthesis)
Fonts	Self-hosted via @fontsource-variable
Styling	Inline styles + CSS custom properties
State	React hooks; no Redux/MobX
Testing	Vitest
Lint	ESLint + react-hooks plugin
CI	GitHub Actions (lint + build on PR)

System diagram



The browser talks directly to Supabase for all CRUD operations and to Vercel for AI calls. No bespoke backend exists.

Frontend architecture

Routing

Folios uses URL-based view selection but does not use React Router. A simple `view` state in `App.jsx` drives which top-level component renders. This keeps the bundle smaller and the navigation logic explicit. URLs encode the active view + selected account; back/forward work via standard browser history.

Code splitting

Every top-level view is lazy-loaded via React.lazy + Suspense:

```
const AccountsView = React.lazy(() =>
import("./views/accounts/AccountsView"));
const GaugeView    = React.lazy(() =>
import("./views/gauge/GaugeView"));
// ...
```

Initial bundle is the shell + Pip + auth. Other views load on demand. This is what makes the post-deploy stale-chunk failure mode possible — and what makes the auto-recovery layer necessary (see `src/main.jsx` and `src/lib/errorLog.js`).

Data layer (hooks)

Every domain entity has a dedicated hook that owns its read + write surface:

- `useAccounts(userId, orgId)` — accounts CRUD
- `useMeetings(userId, accountId, orgId)` — meetings CRUD
- `useItems(userId, accountId, orgId)` — open items CRUD
- `useContacts(userId, accountId, orgId)` — contacts CRUD
- `useCadences(userId, accountId, orgId)` — cadences CRUD

- `useProjects(userId, accountId, orgId, childAccountIds)` — Gauge projects
- `useQuickTasks(userId)` — quick tasks tray
- `usePipAccountState(userId)` — Pip's per-account memory
- `usePipAssignmentHints(userId, accountId)` — Pip's assignment learning
- `usePipCorrections(userId, accountId)` — Pip's correction log
- `useGlossary(userId, orgId, accountId)` — Pip's glossary
- `useErrors(userId, opts)` — error log
- `useActivity(userId, opts)` — audit log
- `useOrg(userId)` — org membership

Each hook:

- Subscribes to a Supabase Realtime channel for its table
- Refetches on realtime change (debounced ~500ms)
- Exposes loading + error states
- Returns CRUD functions that fire optimistic writes via Supabase client (RLS-enforced server-side)

Components stay presentational — no data fetching in component code.

Theme system

Two themes (dark default, light spec'd) implemented via CSS custom properties on `<html data-theme="…">`. The C object in `src/lib/colors.js` exposes `var(--…)` references, so all inline `style={{ background: C.surface }}` consumers re-theme instantly with no remount.

Pre-mount theme application is done by an inline `<script>` in `index.html` to prevent flash-of-wrong-theme.

Animation

Pip's mood-driven animations are CSS-only (keyframes + state classes via `PipStateProvider` context). The animated mark glyphs use a shared rAF engine in `Mark.jsx`. Both respect `prefers-reduced-motion`.

PWA

Service worker generated by Workbox via `vite-plugin-pwa`:

- `skipWaiting: true`, `clientsClaim: true`, `cleanupOutdatedCaches: true`
- Precaches static assets + shell HTML
- Two redundant auto-update paths in `src/main.jsx`:
  - `controllerchange` event listener (the "right" way)
  - Version polling every 3 min + on visibility change (fallback when SW is stuck)
- Both converge on `triggerReload()` with 60-second cooldown to prevent reload loops during mid-deploy edge-flip-flop.

Mobile

- Mobile-first layout with `useBreakpoint()` hook (900px threshold)
- Mobile shell: top header + scrollable content + fixed bottom nav
- All inputs ≥16px font (prevents iOS Safari auto-zoom)
- Safe-area insets respected (`env(safe-area-inset-*)`)
- Sheet-style modal on mobile vs. centered on desktop

Backend architecture

Supabase Postgres

All CRUD goes directly browser → Supabase via the JS client. Auth is handled by Supabase Auth; every request carries the user's JWT.

Row-Level Security

Every user-data table has RLS policies. Example:

```
alter table folio_accounts enable row level security;

create policy "accounts_select_own" on folio_accounts
  for select using (auth.uid() = user_id);

create policy "accounts_insert_own" on folio_accounts
  for insert with check (auth.uid() = user_id);
```

For org-shared tables, the policy expands to org membership lookup.

Realtime

Every domain hook subscribes to its table's realtime channel:

```
supabase.channel("folio_accounts:" + userId)
  .on("postgres_changes", { event: "*", schema: "public",
       table: "folio_accounts", filter: "user_id=eq." + userId },
      debouncedRefetch)
  .subscribe();
```

Debounce prevents thrash; refetch ensures the local cache stays authoritative. Multi-device sync works because every device of the same user subscribes to the same

channel.

Migrations

SQL migrations live in `supabase/*.sql`. Run manually against production via Supabase Dashboard or `psql`. Canonical schema is in `supabase/schema.sql` (kept in sync as the source of truth).

Migration convention:

- Idempotent (`create table if not exists`, `alter table ... add column if not exists`)
- Single-block (the whole migration runs as one SQL statement)
- Safe to re-run

Pip pipeline

```
User input → classifyIntent → mode-specific prompt → Anthropic API →
  ↓                                                       ↓
streaming response →                                tool calls →
  ↓                                                       ↓
UI updates (PipView)                                executeTool →
                                                    Supabase writes
```

Intent classification

`classifyIntent(text)` (in `src/lib/pipIntent.js`) inspects the user's question and routes it to one of:

- **chat** — conversational answer
- **action** — should result in a tool call (add item, set follow-up, etc.)

- **deterministic** — can be answered locally without an API call (e.g., "how many open items do I have?")

Modes

`callPipApi(messages, context, opts)` is the single network entrypoint. `opts.mode` selects the system prompt and context shape:

- `chat` — general Q&A with full context
- `brief` — pre-call brief for one account
- `summary` — meeting summarize with structured plan output (uses 4-block stacked prompt caching)
- `extract` — short-form action extraction from quick notes
- `compress` — distills correction log into lessons_learned paragraph

Tool calls

For action-mode requests, Pip can return tool_use blocks. `executeTool` in `src/lib/pipExecutor.js` dispatches to the right hook function (addItem, setFollowUp, closeItem, etc.). Tool calls are categorized:

- **Frictionless** — auto-execute (e.g., adding a quick task user asked for)
- **Confirm-required** — surfaced as PipActionCard for user approval

Cost optimization

- 4 cache breakpoints in summary mode (system / glossary / roster / items+tasks). Multi-call sessions get ~70% cache hit.
- Trivial draft skip: drafts <100 chars never call the API.
- Per-call telemetry written to `folio_pip_usage`.
- Rate limit: 20 req/min per user, enforced in `/api/pip`.

Streaming

Streaming responses use Anthropic's SSE format, parsed in `src/lib/pipStream.js`. Each chunk fires an `onDelta(text)` callback that progressively updates the UI.

Learning loop (V2 brain)

Three persistent tables feed Pip's learning:

- `pip_correction_log` — every user correction (last 10 read back per summarize call)
- `pip_account_state.lessons_learned` — periodically compressed paragraph distilled from corrections (~every 5 meetings)
- `pip_assignment_hints` — who Pip should route each kind of task to
- `pip_glossary` — user-taught vocabulary

See `docs/ai-governance.md` for the full learning-loop description.

Build & deploy

Local dev

```
npm install
npm run dev    # Vite dev server on :5173
```

`.env.local` carries `VITE_SUPABASE_URL` and `VITE_SUPABASE_PUBLISHABLE_KEY`. Anthropic key lives only in Vercel env vars.

Production build

```
npm run build  # Vite production build to dist/
```

Vercel deploys on every push to the configured production branch (`claude/build-folio-desktop-app-XzvZ5`). Builds are isolated per branch; only the production branch pushes count toward Vercel's deployment limit.

Migrations

Run manually against production Supabase via the Dashboard SQL editor or `psql` . Migrations are idempotent so re-runs are safe.

Secrets

- `ANTHROPIC_API_KEY` — Vercel env var (production + preview)
- `VITE_SUPABASE_URL` and `VITE_SUPABASE_PUBLISHABLE_KEY` — public, shipped to client by design
- Supabase service-role key never used in client code; only in local maintenance scripts run by the developer with their own credentials

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Multi-tenancy readiness

Even at single-user pilot, the architecture supports multi-tenant team mode without restructure:

- `folio_orgs` table for orgs
- `folio_org_members` for user-to-org mapping with role
- Every user-data table carries `org_id` (or is linked via account which carries it)

- RLS policies branch to check org membership for shared resources
- Org-wide assignment hints (account_id = null) so Pip can learn team-level patterns
- Activity log scoped per-org for owners

When team mode lights up, the changes are mostly UI (invite flow, member management, role-based visibility), not schema.

Observability

Client-side error capture

`src/lib/errorLog.js` exposes `logError(type, message, opts)`. Three sources:

- `installGlobalErrorHandlers()` (window.onerror + unhandledrejection)
- `ErrorBoundary.componentDidCatch` for React render errors
- Explicit `logError` calls from network/fetch helpers

Captured errors land in `folio_errors` (RLS-scoped per user). Visible in Settings → Diagnostics.

Self-healing

`looksLikeChunkReload(message, stack)` detects stale-chunk Lazy import failures and triggers `triggerReload()` immediately, with the 60s cooldown. Errors of this type are logged as `chunk_reload` with `resolved: true` so they don't clutter the unresolved list.

Performance

`timed(label, fn)` wraps hot-path operations and logs duration. Currently logs to console; future plan is to push to a perf table.

Cost

`folio_pip_usage` tracks every Pip API call (tokens, mode, cache hit flag, timestamp). Per-user cost dashboard in Settings → Pip Usage.

Key architectural decisions & tradeoffs

"No bespoke backend"

Decision: all CRUD goes browser → Supabase directly. Only AI calls go through a Vercel serverless function.

Tradeoff: Faster development, fewer moving parts, single layer of RLS enforcement (no chance of a custom backend forgetting to apply the right scope). Cost: complex queries that span tables are harder to express as a single RPC; we lean on multi-fetch + client-side joining.

"Inline styles + CSS variables"

Decision: no styled-components, no Tailwind, no CSS modules. Just inline styles consuming CSS custom properties.

Tradeoff: Smaller bundle, no runtime style overhead, instant re-theme via CSS variable swap. Cost: shared style patterns are slightly more verbose; we offset with `src/lib/colors.js` (`C` object) for token reuse.

"Hooks-first data layer"

Decision: every entity has a dedicated hook; components are presentational.

Tradeoff: Clear boundaries, easy to test data logic in isolation, straightforward realtime wiring per table. Cost: components that need multiple entities (e.g., AccountDetail) end up with many hook calls; we accept the verbosity for the clarity.

"Optimistic writes via Supabase client"

Decision: no explicit mutation library (no React Query, no SWR mutations). Writes go through Supabase JS client directly with realtime echoing the update back.

Tradeoff: Simpler mental model, fewer dependencies. Cost: no built-in retry-with-backoff for mutations; we add it manually in `src/lib/net.js` where needed.

"Pip is a thin proxy"

Decision: `/api/pip` is a thin auth + rate-limit + prompt-shape proxy to Anthropic. No conversation state stored server-side; client sends the full message history every call.

Tradeoff: Stateless, easy to scale. Cost: prompt size grows with history; we mitigate with prompt caching + per-mode context shaping.

File layout (top level)

```
/api                    Vercel serverless functions (Pip proxy)
/docs                   Presentation docs (this directory)
/scripts                Local maintenance scripts (seed data, etc.)
/src
  /components           Reusable UI primitives
  /hooks                Domain data hooks (one per entity)
  /layout               App shell (MobileLayout, DesktopLayout)
  /lib                  Cross-cutting modules (colors, pip, errorLog,
...)
  /views                Top-level pages, lazy-loaded
/supabase               SQL migrations and canonical schema
CLAUDE.md               Development context, queue, rules (internal)
vercel.json             Deploy config + cache headers
vite.config.js          Build + PWA config
index.html              Shell HTML + theme bootstrap + global CSS
```

Contact

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